



Master at work: Pete Sampras (in white), like all excellent servers, leapt when he served. But he attempting to land his split step at the instant of his opponent's return, for instance against Andre Agassi (in black), thereby giving himself time to take as many steps as possible towards the oncoming ball rather than just towards the net.

possible towards the oncoming ball rather than just towards the net. In several of the serves of his match against his arch-rival, you can see Sampras quickly adapt to returns that require stopping his progress towards the net and his resulting calm, balanced setup for what could have been an awkward first response to the return.

On shots besides first volleys, Sampras also appears to be trying to land his split step just as Agassi hits the ball, if he has time to do so.

Is it fair to say that tennis is a game of split seconds just as much as — or even more than — it's a game of inches? And if so, why? Split seconds are more important. The reason is simple. A high-level player moving laterally at 15 feet per second, which is not unusual, moves, on average, 1.5 feet every tenth of a second.

So, in this example, a tenth of a second, which is not much time, 'equals' 1.5 feet. Two-tenths of a second 'equals' three feet.

To get an idea of how fast two-tenths of a second is, rhythmically make the 'tsk' sound 11 times (rather than counting, it's easier as two groups of four and one group of three). Adjust the time between the tsks so that if the first tsk occurs when you hit 'start' on a stopwatch, the last tsk occurs at exactly 2.00 seconds. Then, the time between each tsk will then be 0.2 seconds.

The slower the speed of shots and the slower the player is capable of moving, the less important time becomes. Conversely, as pro tennis has sped up this century — with both the players moving faster and the ball travelling faster — time, especially delays in starting movements, becomes more important.

Which physical attributes play key roles in the split step?

For the ideal split step, three physical attributes are important.

First, there is the player's reaction time to visual cues. The opponent hitting the ball and the initial few feet of the ball's travel off the opponent's racquet is a visual cue. For most young humans, athletes or not, their reaction time is 0.2 to 0.25 seconds. Reaction times increase with age but can be shortened with training. World-class tennis players engage in such training, so a reaction time of 0.2 seconds is not difficult to achieve.

Second, there is the player's rate of force development, which is affected by a complex of neural, muscular, and other body components that act together to produce a 'fast twitch'. The most important thing about force development is not understanding the complex interplay of every bodily